

Types of Charts in Power BI & When to Use Them

Power BI provides a variety of **charts and graphs** for data visualization. Choosing the right chart depends on the type of data and the insights you want to communicate.

1. Bar & Column Charts

- ♦ **Used for:** Comparing categorical data across different groups.
- ♦ **Best for:** Showing trends, ranking, and distribution.

Chart Type	Description	When to Use
Clustered Bar Chart	Displays multiple bars for different categories side by side.	Comparing multiple categories with different values.
Stacked Bar Chart	Stacks bars to show the total and contribution of each category.	Showing total values with sub-category breakdown.
100% Stacked Bar Chart	Each bar is scaled to 100%, showing the proportion of each category.	Comparing relative percentage contributions.
Clustered Column Chart	Vertical bars comparing multiple categories.	Comparing multiple categories over time.
Stacked Column Chart	Stacks columns to show category breakdown.	Showing parts of a whole over time.
100% Stacked Column Chart	Shows category proportions within 100% scale.	Comparing relative percentages across categories.

📌 **Example:** Comparing **monthly sales by product category** → Use a **Clustered Column Chart**.

2. Line & Area Charts

- ♦ **Used for:** Showing trends and patterns over time.
- ♦ **Best for:** Time series data and forecasting.

Chart Type	Description	When to Use
Line Chart	Displays data points connected by lines.	Tracking trends over time (e.g., sales growth).
Stacked Area Chart	Filled area under the line to show cumulative value.	Comparing category trends and their total value over time.
100% Stacked Area Chart	Each area scaled to 100%.	Showing proportional trends over time.

📌 **Example:** Tracking **monthly revenue growth over a year** → Use a **Line Chart**.

3. Pie & Donut Charts

- ♦ **Used for:** Showing proportions of a whole.
- ♦ **Best for:** Comparing percentage distribution.

Chart Type	Description	When to Use
Pie Chart	Divides a circle into slices representing percentages.	Comparing a few categories (5 or fewer).
Donut Chart	Similar to a pie chart but with a hole in the center.	Showing category proportions while keeping space for additional insights.

📌 **Example:** Analyzing **market share distribution by brand** → Use a **Pie Chart**.

4. Scatter & Bubble Charts

- ♦ **Used for:** Identifying relationships and distributions between data points.
- ♦ **Best for:** Correlation analysis and outlier detection.

Chart Type	Description	When to Use
------------	-------------	-------------

Scatter Chart	Plots points on an X-Y axis to show relationships.	Identifying correlations between two numerical variables.
Bubble Chart	Similar to a scatter chart but with bubble size representing an extra variable.	Comparing three variables in one chart.

📌 **Example:** Analyzing **sales vs. profit by region** → Use a **Scatter Chart**.

5. Maps & Geospatial Charts

- ♦ **Used for:** Visualizing geographic data.
- ♦ **Best for:** Showing distribution across locations.

Chart Type	Description	When to Use
Map (Basic Map)	Plots data points on a map.	Displaying sales or customer locations.
Filled Map (Choropleth Map)	Shades areas based on values.	Comparing values across regions (e.g., sales per state).
ArcGIS Map	Advanced mapping with layers and analytics.	Detailed geospatial analysis.

📌 **Example:** Mapping **customer locations by country** → Use a **Basic Map**.

6. Waterfall & Funnel Charts

- ♦ **Used for:** Showing contributions and flows in data.
- ♦ **Best for:** Analyzing sequential data.

Chart Type	Description	When to Use
Waterfall Chart	Shows how a starting value changes due to additions/subtractions.	Analyzing profit margins or financial statements.
Funnel Chart	Displays stages in a process and drop-off rates.	Sales funnel analysis (e.g., lead conversion rate).

📌 **Example:** Showing **profit breakdown from revenue to net income** → Use a **Waterfall Chart**.

7. Gauge & KPI Cards

- ♦ **Used for:** Displaying progress towards a goal.
- ♦ **Best for:** Performance tracking.

Chart Type	Description	When to Use
Gauge Chart	Circular chart showing progress toward a target.	Displaying key performance metrics.
KPI Card	Shows a single metric compared to a goal.	Monitoring performance vs. target.

📌 **Example:** Displaying **monthly revenue vs. target** → Use a **KPI Card**.

8. Tables & Matrices

- ♦ **Used for:** Presenting detailed data.
- ♦ **Best for:** When users need raw numbers in tabular format.

Chart Type	Description	When to Use
Table	Displays raw data in rows and columns.	Viewing detailed data records.
Matrix	Like a pivot table, allows grouping and drilling.	Analyzing data with multiple dimensions.

📌 **Example:** Listing **sales data by product and region** → Use a **Matrix**.

9. Treemaps & Heatmaps

- ♦ **Used for:** Showing hierarchical data and intensity-based analysis.
- ♦ **Best for:** Categorization and trend analysis.

Chart Type	Description	When to Use
Treemap	Displays hierarchical data as nested rectangles.	Comparing sub-categories within a category.
Heatmap (Custom Visual)	Uses color intensity to show data distribution.	Highlighting high and low values across a grid.

📌 **Example:** Sales contribution by product category → Use a **Treemap**.

10. Custom Visuals (Marketplace)

- Power BI allows importing **custom visuals** from **Microsoft AppSource**.
- Examples: **Sankey Diagram, Bullet Chart, Timeline Visual, Advanced Gauge**.

📌 **Example:** Visualizing **customer journey flow** → Use a **Sankey Diagram**.

Choosing the Right Chart in Power BI

Goal	Best Chart Type
Compare values across categories	Bar, Column, Treemap
Show trends over time	Line, Area, Column
Show part-to-whole relationships	Pie, Donut, Stacked Bar/Column
Show relationships between variables	Scatter, Bubble
Show progress toward a target	Gauge, KPI Card
Show geographical data	Map, Filled Map
Show a sequential process	Funnel, Waterfall
Show raw data	Table, Matrix

Conclusion

Choosing the right chart in Power BI depends on **the type of data and the insights you want to communicate**. Using appropriate charts helps make reports more **meaningful, insightful, and actionable**.

Would you like help designing a Power BI dashboard? 